

Boyue Du

AI/ML Researcher | NVIDIA Hackathon Winner | IEEE-HKN Mentor
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Education

University of California, Los Angeles (UCLA) <i>Master of Science in Electrical and Computer Engineering</i> Research Areas: Machine Learning, Representation Learning	Los Angeles, CA <i>Sep. 2025 – Jun. 2027 (Expected)</i>
University of Wisconsin–Madison <i>Bachelor of Science in Electrical Engineering GPA: 3.7/4.0</i>	Madison, WI <i>Sep. 2021 – May 2025</i>

Technical Skills

Languages:	Python, Java
AI & Deep Learning:	PyTorch, Transformers, scikit-learn, Multimodal AI, Generative AI, Reinforcement Learning
Systems & Tools:	NumPy, pandas, Linux, CUDA, ROS, Git, REST APIs, LangChain

Selected Research & Projects

Representation Learning of Fatigue-Induced Lower-Limb Muscle Synergies <i>Research Project Lead, Prof. Ma's Lab, Beijing Normal University</i> - Built an end-to-end ML pipeline for 325 trials of 16-channel bilateral EMG, integrating signal filtering, event alignment, normalization, and feature extraction to analyze fatigue-related muscle coordination. - Designed a time-preserving autoencoder (AE) and benchmarked it against NMF using repeated training, Hungarian matching, and leave-one-trial-out validation, achieving 89% spatial similarity and 97.4% reconstruction VAF.	2026 <i>Remote</i>
Decoding Keystrokes from sEMG Signals Using Deep Neural Architectures <i>UCLA</i> Developed CNN-, LSTM-, and Transformer-based models for sEMG keystroke decoding in PyTorch, integrating spectrogram features, data augmentation, CTC-based sequence modeling, and beam-search decoding, achieving a 54% relative reduction in Character Error Rate (CER) versus the baseline.	2026 <i>Los Angeles, CA</i>
Vision-Language-Action Model Evaluation for Autonomous Driving <i>Mobility Lab, UCLA</i> Evaluated NVIDIA Alpamayo-R1 across 1,000 driving clips and built Python pipelines to sweep decoding temperature, top-p, and trajectory-history length using minADE, identifying temperature 0.2 as the best tested configuration with a 3.89 mean minADE.	2026 <i>Los Angeles, CA</i>

Awards

Culture Impact Track Winner – NVIDIA GTC Hackathon <i>NVIDIA, sponsored by HP</i> - Solo-developed <i>Symphonia</i>, an end-to-end multimodal AI application integrating audio processing, LLMs, and interactive visualization for classical music understanding and content generation. - Built and deployed a GPU-accelerated inference pipeline integrating NVIDIA Nemotron and OpenAI APIs for real-time multimodal audio analysis on an edge AI system.	2026 <i>San Jose, CA</i>
Novotny, Donald W. Scholarship, UW–Madison	2024
WV Distinguished Scholarship, UW–Madison	2023, 2024

Additional Experience

Graduate Reader <i>ECE 113 Digital Signal Processing, UCLA</i> Graded homework and exams for a 200+ student course and provided technical feedback.	Mar. 2026 – Jun. 2026 <i>Los Angeles, CA</i>
Mentor, IEEE-Eta Kappa Nu (IEEE-HKN) <i>UCLA</i> Mentored undergraduate engineering students through individualized guidance on academic planning, professional development, and long-term career goals.	Mar. 2026 – Jun. 2026 <i>Los Angeles, CA</i>
Business Developer <i>NaaS Technology</i> Represented the company at CES 2024 and provided technical consultations on mobile and stationary EV charging products.	Nov. 2023 – Jan. 2024 <i>Las Vegas, NV</i>

Certifications

ROCm Certification Associate: Architecture, Programming, and Optimization – AMD Advancing AI
Building LLM Applications with Prompt Engineering – NVIDIA GTC

Publications

L. Song, Z. Yang, J. Guo, **B. Du**, and Z. Pang, “Diversiform propagation of hollow Gaussian beam clusters in non-linear media with self-induced parabolic potentials,” *Journal of Optics*, vol. 22, no. 11, Art. no. 115609, 2020. doi:10.1088/2040-8986/abbb60 .